MIL-D-81465A(AS)
30 July 1971
Superseding
MIL-D-81465(AS)
13 April 1967

MILITARY SPECIFICATION

Detecting Group, Submarine Anomaly (SAD) AN/ASA-64A

This specification has been approved by
Naval Air Systems Command, Department of the Navy

1. SCOPE

*1.1 Scope - The equipment covered by this specification shall be for the automatic recognition and marking of submarine signals detected by Magnetic Anomaly Detection (MAD) equipment. The equipment shall operate in all types of aircraft designated for anti-submarine warfare and equipped with MAD equipment. It shall generate a voltage pulse which can be applied to a computer, coordinate plotter, strip chart recorder, audio oscillator or any other such equipment. The controls for the detector shall be incorporated in the C-7693/ASA-71.

1.2 Classification - The equipment covered by this specification shall consist of the items listed in 6.8.

2. APPLICABLE DOCUMENTS

*2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-R-5857	Relays (electrical (excluding thermal) for Electronic and Communication - type equipment), General Specification for
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

FSC 5895

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MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-S-19500	Semiconductor devices, General Specification for
MIL-C-81511	Connector, Electrical, Circular, High Density, Quick Disconnect, Environmental Resisting General Specification for

Bureau of Naval Weapons

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of
WR-101	Weapons Requirements for Advanced ASW Electronic Systems Part I - Electromagnetic Control Requirements for Advanced ASW Avionics System

STANDARDS

Military

MIL-STD-415	Test Points and Test Facilities for Electronic Systems and Associated Equipment, Design Standard for
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipment)
MIL-STD-471	Maintainability Demonstration
MIL-STD-701	Preferred and Guidance Lists of Semiconductor devices
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-781	Test Levels and Accept/Reject Criteria for Reliability of Non-Expendable Electronic Equipment

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MIL-STD-785 Requirements for Reliability Program (for Systems and Equipments)

MIL-STD-831 Test Reports, Preparation of

OTHER PUBLICATIONS

FED-STD-595 Color

*2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

2.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specification - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

2.4 Procuring Activity - The term "procuring activity" used in this specification shall be defined as the procuring activity and/or its authorized representative.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provisions for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials fulfillment of major design objectives shall be the prime consideration. In doing so the following shall govern:

(1) Microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

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(3) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuits to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must have a Mean-Time-Between-Failure (MTBF) of 5000 hours minimum. (See 6.4)

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be chosen and approved as outlined in AR-5.

3.2.2 Electron Devices - Transistors and diodes shall be chosen and applied in accordance with the requirements of MIL-S-19500. Semi-conductor devices listed in MIL-STD-701 shall be utilized whenever possible.

3.2.3 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of AR-5.

3.2.4 Maintenance Module - A maintenance module is defined as the least complex segment of the contract end article(s) which is capable of being removed and replaced at organizational level maintenance using tools, support equipment and procedures approved for the end article(s). Further, it is the segment to which the Contractor will provide fault detection and isolation techniques consistent with scheduled and unscheduled organizational level maintenance requirements.

3.2.5 Microelectronic Interchangeability - The interchangeability provisions of AR-5 shall govern Microelectronic Interchangeability.

3.3 Design and Construction - The equipment shall conform with the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

*3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 7 pounds.

3.3.2 Reliability

3.3.2.1 Reliability Program - The contractor shall establish and conduct a reliability assurance program in accordance with MIL-STD-785.

3.3.2.2 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 250 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

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3.3.2.3 Operating Life - The equipment shall have total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement and replacement intervals shall be specified by the contractor.

3.3.2.4 Reliability in Mean-Time-Between-Failure (MTBT)- The specified Mean-Time-Between-Failures for the equipment shall be 1000 hours when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2.5 Time Totalizing Meter - The equipment shall contain a time totalizing meter in accordance with Specification MIL-M-7793.

3.3.3 Cabling and Connections.

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification WR-101 (Part I) and MIL-C-81511.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification WR-101 (Part I). The external wiring shall be unshielded, except that a minimum number of individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Interchangeability of Reordered Equipment - For reordered equipment, interchangeability shall exist between units and all replaceable assemblies, subassemblies and parts of a designated model of any previously manufactured equipment, supplies or designated by the procuring activity.

3.3.5 Interference Control - The equipment shall conform with all of the requirements of WR-101 (Part I) for design, construction and workmanship. A Control Plan will be prepared in accordance with WR-101 (Part I).

3.3.6 Maintainability.

3.3.6.1 Maintainability Program - The contractor shall establish and conduct a maintainability program in accordance with MIL-STD-470 and MIL-STD-471. All maintenance actions shall be capable of being performed by a technician with a comprehensive ability level as follows:

(a)	Civilian education:	High School graduate or equivalent.
(b)	Technical training:	Navy, Class A Technical school appropriate to the maintenance task.
(c)	Experience:	Two years of technical experience in addition to technical training

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3.3.6.1.1 Readiness Measurement Requirements - Suitable features shall be incorporated in the equipment to provide the operator, while airborne and while on the ground, with GO-NO-GO indication of equipment readiness. These features may be in conjunction with, but not necessarily part of the fault isolation features.

3.3.6.1.1.1 Mode of Operation - The operation of the performance/readiness test feature shall be accomplished by a manually energized test mode of operation.

3.3.6.1.2 Maintainability Requirements

3.3.6.1.2.1 Operational Maintainability Requirements - The time required to perform unscheduled maintenance actions shall be composed of:

1. Recognition of a fault
2. Isolation of the fault to a maintenance module
3. Maintenance module replacement
4. Retest for system readiness

Mean Repair Action Time is the summation of all maintenance downtimes during a given period divided by the number of maintenance tasks performed during the same period. For organizational maintenance this number shall not exceed 30 minutes. The maximum time required to accomplish any single maintenance action at organizational level will not exceed 60 minutes.

3.3.6.1.2.2 Packaging Requirements - The equipment shall be packaged in modular form so that, to the maximum extent practical, all subassemblies shall be of the plug-in type and should be easily removable without special tools, for maintenance. If a special tool is required for removal, it shall be provided and attached to the equipment.

3.3.6.1.2.3 Module Commonality - The number of different modules will be kept to an absolute minimum. This standardization is mandatory to ease the testing and spares requirements at all levels of maintenance.

3.3.6.1.2.4 Module Accessibility - Module arrangement, will be such that access to each module does not require the removal of adjacent modules or parts other than access panels.

3.3.6.1.2.5 Keying - Modules and connectors shall be keyed to prevent the insertion of a connector or module into an improper location within or on the equipment.

3.3.6.1.2.6 Part Replacement - Part replacement on the analog modules shall be possible by skilled technicians working in a designated service area such as intermediate maintenance activity. Use of potting compounds, excessive moisture/fungus proofing compounds or any other substance to mechanically support piece parts is forbidden. Moisture/fungus proofing compounds are permitted provided:

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(1) A 30 mil thickness is not exceeded.
(2) Reference designators and/or part identification remain clearly legible.
(3) Compliance with paragraph 3.2.1.1.9 of MIL-E-5400 is maintained.
(4) Part placement is not impeded.

3.3.6.1.2.7 Non-repairable Assemblies - All designs for non-repairable assemblies shall be submitted to the procuring activity, for approval, prior to release for fabrication.

3.3.6.1.2.8 Adjustments - Procuring activity approval is required for design and use of all periodic adjustment, alignment, or calibration functions except for those specified herein. Adjustments shall be capable of being made in the aircraft using simple hand tools and/or general and standard test equipment. Adjustments shall be identified and accessible without a requirement to remove or relocate the module.

3.3.6.1.2.9 Access Panels - Access panels shall be retained by quick disconnect fasteners, i.e., hand operated slide, dzus type, etc.

3.3.6.1.2.10 Service Access - Devices requiring removal periodically for inspection and/or servicing shall be retained by quick disconnect fasteners, i.e., hand operated slide, dzus type, etc.

3.3.6.1.2.11 Mounting Devices - Connectors, receptacles, fuses, circuit breakers or any other electronic parts, shall not be a part of any mounting device associated with the equipment.

3.3.6.1.3 Fault Isolation

3.3.6.1.3.1 Fault Isolation - In conjunction with but not necessarily as part of the Performance/Readiness testing features, there shall be features incorporated which will permit fault isolation to the module level. These features shall include the necessary switching arrangements with a GO-NO-GO readout device that will clearly indicate whether or not the function selected is within minimum performance standards. If a built-in readout device is not feasible, means shall be provided to use general or standard test equipment, i.e., Oscilloscope, VOM, etc.

3.3.6.1.3.2 Diagnostic Ability - If feasible, the equipment shall be logically and electrically designed to facilitate the location of malfunctioning modules through diagnostic programming in conjunction with the aircraft's computer and/or other installed electronic equipment. Test points shall be made available at the front of the equipment to assist diagnosis of malfunctions.

3.3.6.1.4 Test Points

3.3.6.1.4.1 Test Point Requirements - All test points shall be in accordance with MIL-STD-415 and shall accommodate standard Navy approved test equipment, i.e., multimeter or oscilloscope probes. In the event test points cannot accommodate standard probes, use of adapters will be permitted. These adapters are to be provided and secured to the set.

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*3.3.6.1.4.2 Power Supply Test Points - Accessible external individual test receptacles shall be provided on the equipment to measure the power supply output voltages. The receptacles shall be identified as to the voltage normally present at the test point.

3.3.6.1.4.3 Chassis Test Points - Test points and built-in test facilities shall be included in the design of the equipment and shall be in accordance with MIL-STD-415. Test points shall be chosen to provide a straightforward, logical step-by-step trouble shooting sequence, as well as provide a single end-to-end performance check.

3.3.6.2 Maintainability Assurance - Testing to demonstrate the achievement of the maintainability requirement specified shall be in accordance with paragraph 4.5, when required by the contract.

3.3.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment shall be in accordance with Specification MIL-N-18307.

3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

Temperature	Room Ambient (25 $\pm$ 5°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Volt.	115 $\pm$ 1.0 VAC 400 CPS $\pm$ 1%

3.3.9 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or combination of these conditions as specified in Specification MIL-E-5400 for Class 1A equipment, except as modified herein:

3.3.9.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve IV of Specification MIL-E-5400.

3.3.9.2 Altitude - The equipment shall operate satisfactorily at all altitudes from sea level to 20,000 feet, and no permanent damage to the operation of the equipment shall result on transporting in an operating or in a non-operating condition at an altitude up to 40,000 feet. Satisfactory operation shall resume immediately when pressure equivalent to 20,000 feet altitude or lower is reached.

3.3.10 Warm-up Time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed 1 minute under standard conditions and 3 minutes under extreme service conditions. Extreme service conditions shall be defined as one or any combination of conditions as specified in MIL-E-5400.

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3.3.11 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as defined herein. The equipment shall operate within the bounds of limits 2 and 3 of Figure 3 of MIL-STD-704. Degraded operation may occur when the input voltage exceeds the above limits but remains within limits 1 and 4 of Figure 3 of MIL-STD-704. However, no damage shall be suffered by the equipment under transient conditions defined herein. Normal operation shall automatically be resumed upon return of the input voltage to levels within limits 2 and 3. The power required shall not exceed the following specified amounts: (The use of 28 VDC aircraft power in this equipment is prohibited).

(1) AC Power (single phase) 115 V, Category B, 50 VA.

(2) AC Power (single phase) 28 V, Category B, 10 VA (for panel lamps only).

3.3.12 Protective Devices - Circuit breakers shall be used in lieu of fuses whenever possible. Fuses, if used, shall be of approved type. Fuse holders and/or circuit breakers shall be accessible from the front of the equipment plainly marked as to rating and size and shall be of the fault indicating type and/or have access for quick voltage checks.

3.4 Performance - Unless otherwise specified, values set forth to establish satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The equipment shall recognize and mark visually and electrically, MAD submarine type signatures in the presence of geologic, maneuver, geomagnetic and equipment noise as generated by the MAD magnetometer system. This shall be done by (1) selective and passive rejection filtering, (2) short-term integration of signal and noise, and (3) electronic correlation with the maneuvers of the aircraft. The equipment shall receive input signals and noise from the AN/ASQ-8, AN/ASQ-10, AN/ASQ-10A, or AN/ASQ-81. The output of the equipment shall be a voltage (see 3.5) whose amplitude is proportional to the probability of a submarine contact. Voltage outputs may be displayed on a MAD strip chart pen recorder (RO-32/ASQ) and/or serve as input to an integrated ASW system computer, coordinate plotter, audio oscillator or other such storage and warning devices.

*3.4.2 Input Signal Characteristics - The input impedance of the recognition channel shall be greater than 0.5 Megohm, and the input impedance of the inhibition channel shall be greater than 0.150 Megohms in the midband. The input signal to the recognition channel shall be analog and shall not exceed 25V peak to peak. The common mode DC voltage shall not exceed + 300V DC. The input signal to the inhibition channel shall be single ended analog and shall be 7-1/2 volt (± 10%) for ± 25° of aircraft maneuver. The input signal to the inhibitions channel shall contain not more than 1.5V peak to peak of 400 cps aircraft power interference.

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3.4.3 Performance Requirement - The equipment shall detect a MAD submarine type signal in the presence of noise, with a signal to noise amplitude ratio as low as 1.75 : 1. The equipment shall also inhibit maneuver noise false alarms. The Mark indication detection shall occur before or during the longest slope of the signal. Maneuver false alarm inhibition shall cause no inhibition of submarine signal recognition, except when a maneuver noise event occurs simultaneously with a submarine contact.

3.4.4 Frequency Requirements - The equipment shall operate satisfactorily in the MAD real-time frequency range from 0.02 cps to 5 cps within the limitations set by the noise rejection filter of paragraph 3.5.1.4.(2).

3.5 Detail Requirements - The detail requirements of the items listed in 6.8 are defined below.

3.5.1 Indicator, Magnetic Variation, ID-1559/ASA-64A - This unit shall meet the following requirements.

3.5.1.1 Function - This unit shall incorporate all the elements necessary for the recognition and marking of MAD submarine type signals in the presence of background noise as detected by the MAD magnetometer system. The unit shall provide the filter circuits for the rejection of quasisinusoidal geologic and equipment background noise and the switching circuits for rejection of maneuver noise by time correlation with the maneuvers of the aircraft. The voltage pulse, indicating the occurrence of a submarine type signal, shall occur only when the output voltage of the short-term sampling integrator exceeds a preset threshold voltage and the wing dip rate does not exceed a minimum value. The amplitude of the voltage pulse shall be proportional to the difference between the threshold voltage and integrated voltage. The unit shall contain a connector to remote controls and indicators used in flight.

*3.5.1.2 Form Factor - The case enclosing the unit shall not exceed 5.887 inches in width, 6.20 inches in height and 7.5 inches in depth.

*3.5.1.3 Weight and Mounting - The weight of the unit shall not exceed 7 pounds. The unit shall be suitable for hard mounting.

3.5.1.4 Contents - The unit shall contain the following circuits and/or subassemblies; but not be limited to these circuits only.

(1) Signal Reference Converter - The Signal Reference Converter shall convert the floating magnetometer signal output as described in para. 3.4.2, to a grounded signal output.

(2) Noise Rejection Filters - There shall be two Noise Rejection Filters which shall reject the quasisinusoidal geologic and equipment background noise. Each filter shall be a passive twin-T rejection filter. One shall provide 45 db $\pm$ 10% attenuation at 0.023 cps and the other shall provide 30 db $\pm$ 10% attenuation at 0.9 cps. Both shall have a Q of 0.5. The filters shall be isolated from each other by an isolation amplifier.

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(3) Active Full-Wave Rectifier - The Active Full-Wave Rectifier shall accept the signal from the Noise Rejection Filters and full-wave rectify it. The output of the Active Full-Wave Rectifier shall be impressed upon the Sampling Integrator. The gain of the active Full-Wave Rectifier shall be adjustable.

(4) Active Sampling Integrator - The Active Sampling Integrator shall integrate the output of the Active Full-Wave Rectifier (3) for a 2.5 second ± 10% period. The Sampling Integrator output shall then return to zero volts and immediately integrate again for 2.5 seconds and continue this operation.

(5) Threshold Detector - The Threshold Detector shall accept the voltages from the Active Sampling Integrator. The output of the Threshold Detector shall be proportional to the difference of the output voltage from the Sampling Integrator and a preset DC voltage threshold level. The threshold level shall be adjustable in flight by a control on the C-7693/ASA-71.

(6) Output Amplifier - The output amplifier shall bias the output signal of the threshold detector to a minus 7.5 volt ± 10% level and shall provide a positive going mark signal. The output amplifier shall be capable of producing a maximum signal of + 10 volt into a 5 K load.

*(7) Differentiator - Lowpass Filter - The differentiator lowpass filter shall accept the inhibition input signal as described in para. 3.4.2, and differentiate this signal up to 0.5 cps. The lowpass filter shall cut off signals with a frequency higher than 2 cps and shall adequately suppress 400 cps aircraft power interference. The differentiator lowpass filter shall not create a time delay of more than 400 msecs in the inhibiting signal.

(8) Active Full Wave Rectifier - The active full wave rectifier shall accept the signal from the Differentiator - Lowpass Filter (7) and shall full wave rectify it. The active full wave rectifier shall be capable of providing gain.

(9) Maneuver Rate Detector - The maneuver rate detector shall accept the signal from the full wave rectifier (8) and shall only produce a signal if the maneuver rate exceeds a preset absolute value (± 10%).

*(10) Timing Circuit - The timing circuit shall accept the signals from the maneuver rate detector (9), the Threshold Detector (5) and the Multiple Spike Suppressor (11). The timing circuit shall short the input to the output amplifier (6) when the maneuver rate exceeds a preset absolute value, or when triggered by the Multiple Spike Suppressor (11). As soon as the maneuver rate drops below this value, inhibition shall continue for a short fixed (and adjustable) time period, T. If within this time period a signal is present at the output of Threshold Detector (5), the inhibition time shall be extended for an additional period, T. Similiarly, if within this additional period T, another signal is present at the output of Threshold Detector (5), the inhibition time shall

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again be extended for yet another additional period, T, and so on. If no signal is present at the output of the Threshold Detector (5) during an inhibition period, the integrator reset circuit (12) shall be activated and the short on the output amplifier (6) shall be removed.

*(11) Multiple Spike Suppressor - The Multiple Spike Suppressor shall accept a signal from the output amplifier (6). Whenever this signal exceeds 4.5 volts peak excursion (± 10%) a trigger pulse shall be produced which then starts an inhibition cycle of timing circuit (10).

*(12) Integrator Reset Circuit - The integrator reset circuit shall automatically reset the integrator to a potential not exceeding 100 mV, every 2.5 sec (± 10%). The integrator reset circuit shall also accept the outputs from the timing circuit (10) and short the sampling integrator for a short period at the end of the inhibition cycle and prior to the release of the short at the input of the output amplifier (6)

*(13) Relay Circuit - The relay circuit shall provide a contact closure (for the inhibit indicator on the control console) for the time period that the timing circuit (10) produces an inhibit signal. The relay shall be in accordance with MIL-R-5757/13.

3.5.1.5 Adjustments - The indicator, magnetic variation contain the following internal controls:

(1) Gain Control - A potentiometer shall be provided to permit the adjustment of the gain of the active full wave rectifier (3) in the recognition channel, to provide a coarse threshold adjustment. It shall be a screw driver adjustment.

(2) Maneuver Rate - A potentiometer shall be provided to permit the adjustment of the minimum maneuver rate which will produce an output signal to the timing circuit (10). This rate shall be adjustable between 1 degree per second and 35 degrees per second. It shall be a screw driver adjustment.

*(3) Inhibition Time - A potentiometer shall be provided to permit the adjustment of the fixed time period T of timing circuit (10). This time period shall be adjustable between 1 and 10 seconds. It shall be a screw driver adjustment.

*3.5.1.6 Electrical Connections - Electrical connections shall be provided for cabling to the Control Console, C-7693.ASA-71; the MAD set; and aircraft AC power.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the procuring activity. The procuring activity reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction (First Article) Tests
(2) Initial Production Tests
(3) Acceptance Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the procuring activity.

4.2.1 Preproduction (First Article) Test Data - All required test data collected in the conduct of these tests shall be submitted to the procuring activity for review and approval. This data shall include but not be limited to a list of all electrical and electronic parts giving:

(1) Specified voltage, current and temperature ratings
(2) Applied voltage and current
(3) Ambient and surface temperature of temperature critical parts. These data shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include the following:

(1) Environmental tests in accordance with MIL-T-5422
(2) Interference tests in accordance with WR-101, Part I
(3) Shakedown Tests
(4) Power Input Test

4.2.2.1 Shakedown Test - Each preproduction equipment shall be subjected to a shakedown test of 100 continuous hours of failure-free operation. The test shall be conducted in accordance with MIL-STD-781, using test level E and paragraph 5.2.3.2. Failures encountered in the test will be the subject of detailed failure analysis and corrective action. The corrective action loop shall document the assignable cause of failure and the action taken. Corrective actions shall be approved by the procuring activity prior to implementation. The test will then be started from time zero. The performance characteristics to be measured shall consist of a series of functional tests, exercising all portions of the equipment in a manner as stringent as that required during its ultimate application.

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4.2.2.2 Power Input Test - Each equipment shall be subjected to a range of input voltage and frequencies as specified herein.

(1) The input voltage shall be varied within the limits of Figure 3 of MIL-STD-704 as defined in paragraph 3.3.11. The input frequency shall be varied between limits two and three of Figure 5 of MIL-STD-704. The equipment shall perform as specified in paragraph 3.4.

(2) It shall be demonstrated that input voltage transients greater than those specified above shall not cause damage to the equipment and when the voltage returns within the specified limits, normal operation shall automatically resume. (See para. 3.3.11)

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests (See para. 4.2.2). No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk.

4.2.4 Production Equipments - Equipments supplied under the contract shall, in all respects, including design, construction, workmanship, performance, and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection of equipment already accepted by the procuring activity, and it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment shall be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample and that it meets all applicable requirements, including design, construction, workmanship, performance, and quality.

4.3.2 Accessory Materials - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory materials and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for the deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

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4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the services of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Test
(2) Reliability Assurance Test
(3) Special Test
(4) Sampling Test

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of materials, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operational Test
(3) Manufacturing Run-In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run-In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

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Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation.

Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. The input power voltage and frequency shall also be cycled periodically between the specified tolerances. The input voltage transient test specified in paragraph 4.2.2.2 shall be conducted. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Sampling Tests of 4.4.3 are deleted by contractual action).

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

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(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run-in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action proposed.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted as required by MIL-STD-781. Equipments selected for Reliability Assurance Tests shall first have passed the Individual Tests.

4.4.3.1 Reliability Qualification Tests

4.4.3.1.1 Test Level - Test Level E of MIL-STD-781 shall be used with temperature cycling per Figure I of MIL-STD-781.

4.4.3.1.2 Accept - Reject Criteria - Test plan IV of MIL-STD-781 shall be used to determine the accept-reject criteria.

4.4.3.1.3 Performance Characteristics - The performance characteristics to be measured shall be as specified in paragraph 6.9.

4.4.3.1.4 Failure Criteria - In addition to the requirements of MIL-STD-781, the following requirements shall be used to determine when a failure has occurred during the test:

(1) Whenever performance characteristics fall below the acceptance requirements (paragraph 4.4.3.1.3) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

4.4.3.2 Reliability Sampling Tests

4.4.3.2.1 Test Level - Test Level E of MIL-STD-781 shall be used with temperature cycling per Figure I of MIL-STD-781.

4.4.3.2.2 All Equipment Test - Each equipment shall be tested 100 hours. In addition to the 100 hours, a burn-in period may be used at the option of the contractor. If used, the burn-in must be the same for each equipment, in time and severity; deviations, if any, must be approved in advance by the procuring activity. To determine whether the MTBF is being met at any time during the contract, the operation test hours and the failures thereon (not counting burn-in failures or burn-in operation time) shall be totaled and the results compared with the reject line of Test Plan V of MIL-STD-781. (Extend the line as necessary to accommodate the data). These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month, the test results shall be sent to the procuring activity and to the Naval

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Air Systems Command, Attention: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan V curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists, pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc, so that the entire compilation will show other than a reject decision.

4.4.3.2.3 Performance Characteristics - The performance characteristics to be measured shall be as specified in paragraph 6.9.

4.4.3.2.4 Failure Criteria - Failure criteria shall be as specified in 4.4.3.1.4

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipment previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity).

*4.4.4.2 Shakedown Tests - Each of the production equipments submitted for acceptance testing prior to approval of the Reliability Qualification Tests of 4.4.3 shall be subjected to the shakedown tests specified in 4.2.2.1. Upon satisfactory completion of the Reliability Qualification Tests the shakedown tests will be discontinued.

4.4.4.3 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure -Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

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(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Maintainability Demonstration - A Maintainability Demonstration shall be conducted per Method I, Test Plan (b_1) of Table II of MIL-STD-471 and as follows to verify that requirements of 3.3.6.1.2.1 are satisfied.

4.5.1 Test Conditions - One equipment shall be used for this test. The contractor shall select any one of the equipments that is available. All dust covers shall be in place on the equipment prior to the initiation of this test. Before beginning the test, the equipment shall be checked to determine that it is operating normally. Test Equipment, tools, spare parts, diagnostic programs, and documentation shall be available for the test.

4.5.2 Repair Actions - A repair action shall be as follows:

(1) Fault Isolation
(2) Repair
(3) Equipment Checkout

4.5.3 Simulated Failures - The failures shall be randomly introduced, one at a time, throughout the equipment. The points of failures shall be approved by the procuring activity prior to the initiation of the test.

4.5.3.1 Inducing Failures - Single failures shall be induced into the equipment by substituting modules that are known to be defective or, where defective modules are not available, simulated failures will be induced by applying grounds at test jacks or on extended modules. Any failure induced for demonstration purposes will consist of a continuous malfunction.

4.5.4 Repair Time Measurements - The time required for the technician to perform each repair action, including the use of technical manuals and support equipment, shall be measured at the time the technician is made aware that a malfunction exists and ending at the time the equipment is restored to normal operation. Any delay, such as waiting for parts, obtaining test equipment, or administrative delays shall be excluded.

4.6 Test Procedures - The procedures used for conducting all tests required by this specification shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring

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activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to preproduction, reliability assurance and/or maintainability tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

4.10 Test Data Reporting - MIL-STD-831 shall be used as a guide in the preparation of test reports.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

*6.1 Intended Use - The equipment covered by this specification will be used in ASW Aircraft in conjunction with Control Console C-7693/ASA-71, to increase the probability of detection of submarine type anomaly signals by automatic processing of these signals and by shorting the "MAD" signal to ground during aircraft maneuvers of sufficient rate and amplitude to reduce false signals due to the MAD signal produced by the Maneuver.

6.2 Test Values - Normal and limiting values of performance data shall be determined at the standard conditions specified in 3.3.8. These data are to be used in testing the equipment at installation points for compliance with minimum acceptance standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of specific functions beyond the requirements of this specification, are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can

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be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 Non-Repairable Subassemblies - As a general rule, non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts, tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-To-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

*6.5 Type Designation - The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.6 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and data of this specification.
(2) Selection of applicable levels of packaging and packing (see 5.1).

6.7 Asterisk - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the procuring activity assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

*6.8 Items of Equipment - The equipment covered by this specification shall consist of the following items:

Item	Type Designation	Appl. Para.
Indicator, Magnetic Variation	ID-1559/ASA-64A	3.5.1

*6.8.1 Associated Equipment - The equipment shall operate with the following associated equipment:

C-7693/ASA-71,
AN/ASQ-8, AN/ASQ-10, AN/ASQ-10A,
AN/ASQ-81 (Inboard or Towed); RO-32/ASQ;
MIL-D-81347 Data Processing System

6.9 Performance Characteristics to be Measured - Performance characteristics to be measured, to insure that the equipment complies with all requirements of 3.5 (Detail Requirements) and all operational requirements, shall be submitted to the procuring activity for approval.

Project Number 5895-N033

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-D-81465A(AS) DETECTING GROUP, SUBMARINE ANOMALY (SAD) AN/ASA-64A

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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